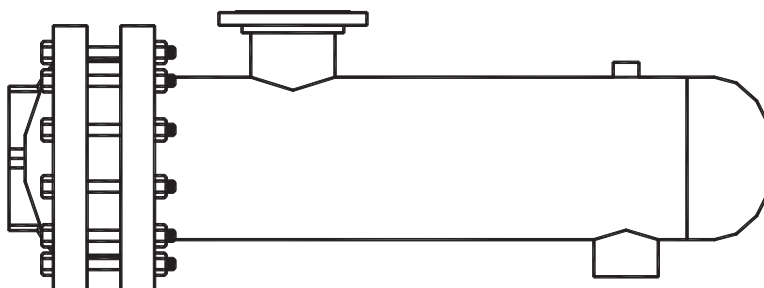




OPERATING & MAINTENANCE INFORMATION FOR HEAT EXCHANGERS



Shell and Tube Heat Exchanger - U-tube style

The THRUSH Heat Exchanger has been carefully assembled and factory tested to provide years of trouble-free service. In order to assure the service intended, the following information is provided to enable proper installation, operation and maintenance of the Thrush Heat Exchanger.

LOCATION

The heat exchanger should be located in a clean, open area, where it is easily accessible for inspection, service and repair. Allowance should be made for the clearance required to remove the tube bundle.

PIPING

Long radius elbows should be used in place of standard elbows wherever possible, because of their superior flow characteristics. Liberal use of shut off valves is recommended to enable inspection and service.

INSTALLATION

1. To move or lift the heat exchanger, place a sling around the unit's inlet and outlet connections. Verify that all equipment, including the sling, is certified to handle the total weight of the heat exchanger. Carefully lift to piping site. Avoid slinging at the bolted gasket head joint area.
2. Install unit level and square so pipe connections can be made without force. During installation and operation, unit should be adequately supported to prevent settling that could cause piping strains.
3. Provide sufficient clearance at channel end for removal and replacement of tube bundle. (Minimum clearance = one length of the shell)
4. Provide positive shutoff valves and by-pass to permit both shell and tubeside to be shut off for inspection and service.
5. After installation and prior to startup all head flange bolts should be re-tightened.

OPERATION

Start up - Open cold side first; then start hot side fluid slowly to gradually bring unit to operating condition. Bring unit from ambient temperature up to operating temperature gradually. Do not introduce hot or cold fluids suddenly. This could damage the unit and void the warranty.

Shut down - Always close off flow of hot fluid first.

Important - Never admit hot fluid suddenly when unit is empty or cold. Do not shock with cold water when unit is hot.

Gasket Creep is inherent to most gasket joints, and retorquing is required. The greater the operating temperature and pressure the greater the problem can become. It is imperative that the head bolts be torqued after installation, after initial startup, and inspected seasonally to be sure the bolts are torqued correctly.

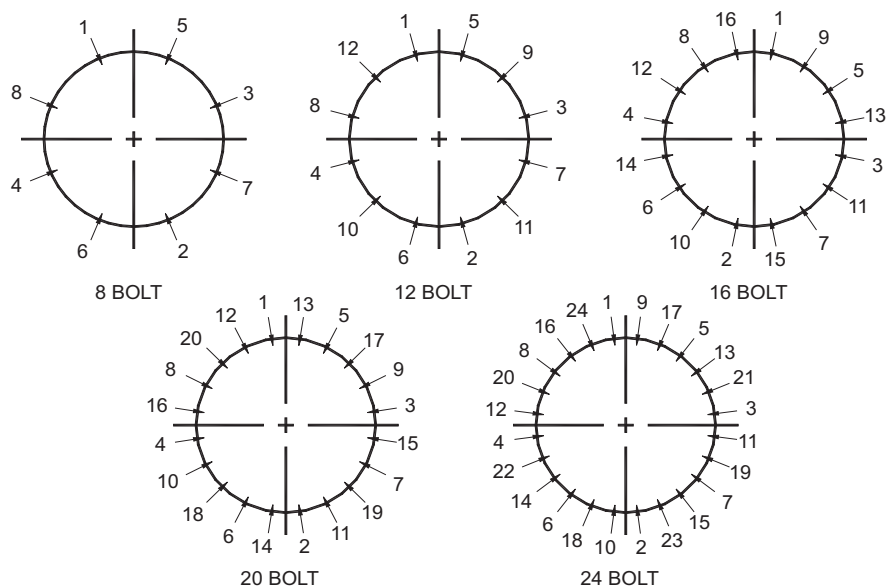
The bolts should be torqued incrementally to 30%, 60%, and then to 100% of the appropriate value in Figure 1. They should be torqued in the sequential order of the appropriate pattern in Figure 2.

Figure 1: Bolt size and Torque requirements

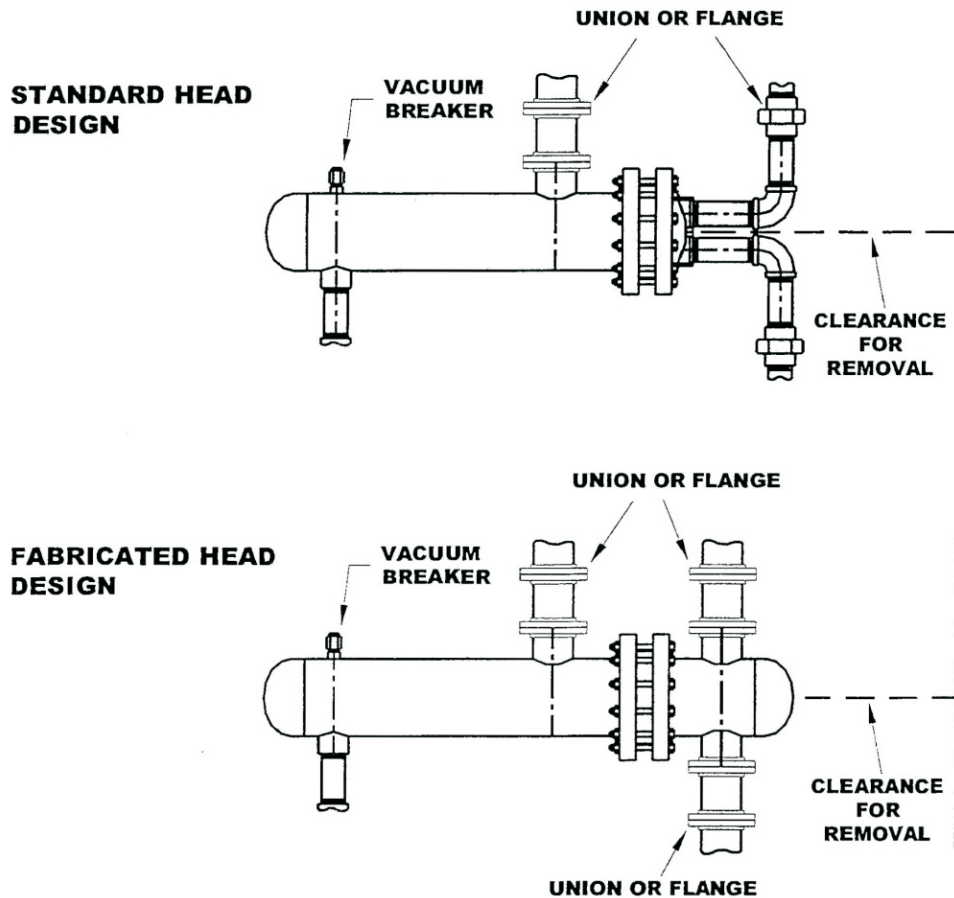
Heat Exchanger Size	Bolt Size	Number of Bolts	Torque Ft/Lb. 150# Flgs.
4"	.62	8	30
6"	.75	8	50
8"	.75	8	50
10"	.87	12	80
12"	.87	12	80
14"	1.00	12	123
16"	1.00	16	123
18"	1.12	16	195
20"	1.12	20	195
22"	1.25	20	273
24"	1.25	20	273
26"	1.25	24	273

Heat Exchanger Size	Bolt Size	Number of Bolts	Torque Ft./Lb. 300# Flgs.
4"	.75	8	100
6"	.75	12	100
8"	.87	12	160
10"	1.00	16	245
12"	1.12	16	390
14"	1.12	20	390
16"	1.25	20	545
18"	1.25	24	545
20"	1.25	24	545

Figure 2: Sequential Order for Torquing Flange Bolts



TYPICAL PIPING

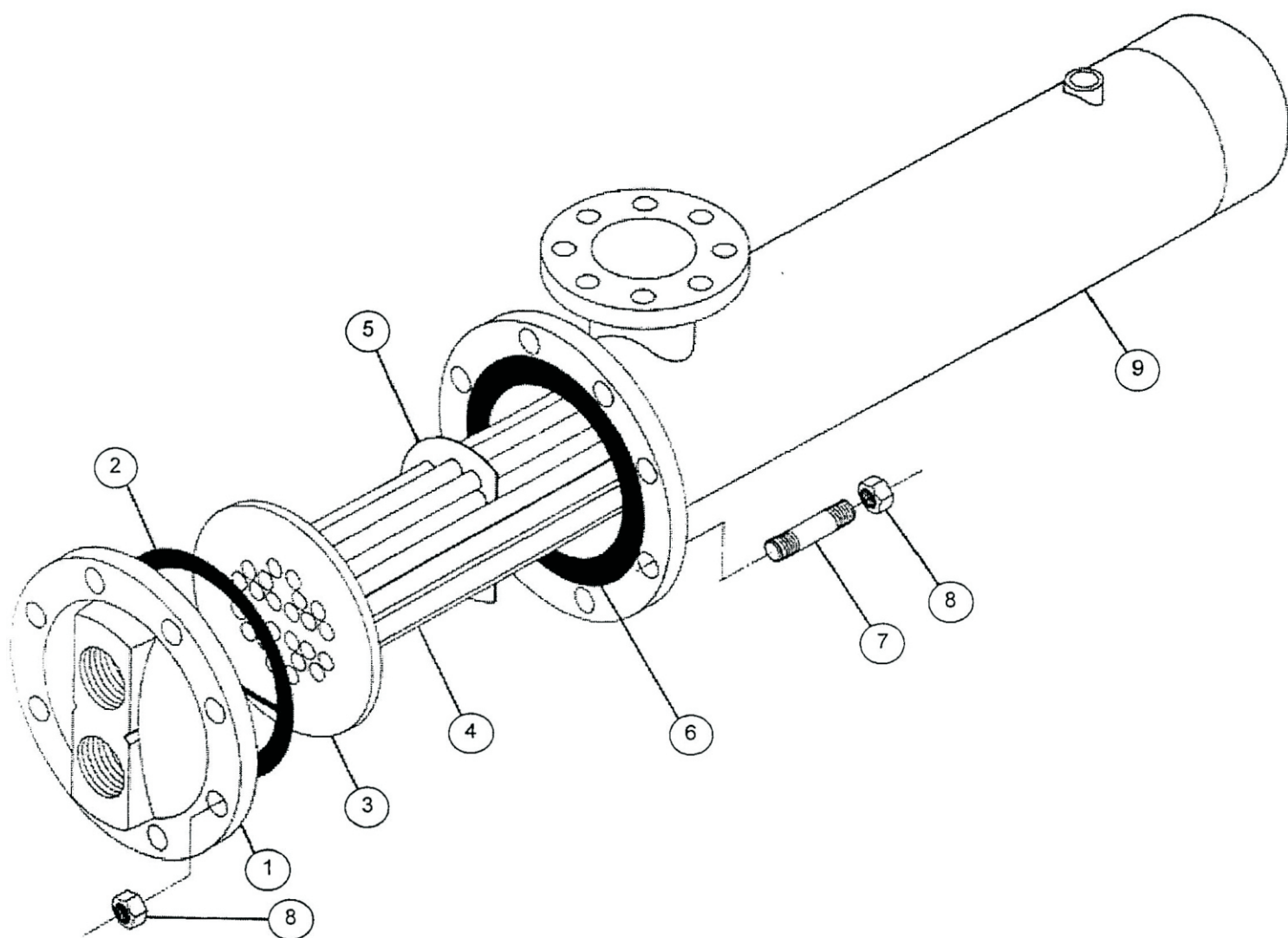


NOTE: Loosen unions/flanges on tubeside heads prior to retorquing head bolts.

MAINTENANCE

Thrush Co., Inc. recommends regular cleaning of the heat exchanger tube bundles. The bundle should be removed periodically and any scale or deposits should be cleaned from the tube bundle surface. Frequency of cleaning is dependent upon water quality and is the responsibility of the Owner. Please note that new gaskets are required for start up each time the unit is disassembled.

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Parts

1. Cast Iron Head
2. Head Gasket
3. Tubesheet
4. Copper Tubes
5. Baffle
6. Shell Gasket
7. Stud Bolt
8. Nut
9. Shell